DTC P243C:00 [SKYACTIV-D 1.5]

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DTC P243C: 00	**Abnormal diesel particulate filter regeneration interval short distance**
DETECTION CONDITION	• The following conditions are met **3 times**: — Drive distance from previous diesel particulate filter regeneration to next diesel particulate filter regeneration is **50 km {31 mile} or less** — PM target accumulation amount is high relative to actual PM regeneration amount **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA is not available. • Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Not applicable
POSSIBLE CAUSE	• Exhaust gas pressure sensor No.2 malfunction • Difference in pressure between exhaust gas pressure before and after passing diesel particulate filter is high • Soot cannot be removed by compulsory diesel particulate filter regeneration • Soot generation amount is high due to abnormal engine combustion • Fuel injector injection amount correction is not completed. • Fuel injector malfunction • Fuel line clogged • Supply pump malfunction • HP-EGR control valve malfunction • Intake shutter valve malfunction • Turbocharger with variable turbine geometry malfunction • PCM malfunction

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	**If DTC P2459 is stored** • Go to the next step. **If DTC other than P2459 is stored** • Repair the malfunctioning location according to the applicable DTC troubleshooting, then go to the next step. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**VERIFY PM ACCUMULATION AMOUNT IN DIESEL PARTICULATE FILTER** • Access the following PIDs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — PM_ACC_DSD — PM_GEN • Start the engine and race it at **3,000 rpm** until the PM_ACC_DSD monitored value stabilizes. • Is the PM_ACC_DSD monitored value twice that of the PM_GEN monitored value or more?	Yes	Go to the next step.
		No	Go to Step 6.

STEP	INSPECTION		ACTION
5	**INSPECT EXHAUST GAS PRESSURE SENSOR NO.2** • Inspect the exhaust gas pressure sensor No.2. (See EXHAUST GAS PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the exhaust gas pressure sensor No.2, then go to Step 13. (See EXHAUST GAS PRESSURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
6	**VERIFY PM ACCUMULATION AMOUNT IN DIESEL PARTICULATE FILTER AFTER COMPULSORY DIESEL PARTICULATE FILTER REGENERATION** • Perform the "COMPULSORY DIESEL PARTICULATE FILTER REGENERATION". (See COMPULSORY DIESEL PARTICULATE FILTER REGENERATION [SKYACTIV-D 1.5].) • Access the PM_ACC_DSD PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the PM_ACC_DSD PID value **approx. 0**?	Yes	Go to the next step.
		No	Compulsory diesel particulate filter regeneration is completed: • Replace the catalytic converter (diesel particulate filter), then go to Step 13. (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
7	**PERFORM FUEL INJECTOR INJECTION AMOUNT CORRECTION** • Perform the "FUEL INJECTOR INJECTION AMOUNT CORRECTION". (See FUEL INJECTOR INJECTION AMOUNT CORRECTION [SKYACTIV-D 1.5].) • Is the fuel injection amount learning completed?	Yes	Go to Step 9.
		No	Verify that the implementation conditions for fuel injection amount learning are met, and re-implement the fuel injection amount learning. • Fuel injection amount learning is completed: — Go to Step 9. • Fuel injection amount learning cannot be implemented: — Go to the next step.
8	**VERIFY IF FUEL INJECTION PRESSURE MALFUNCTION IS POSSIBLE CAUSE** • Inspect the supply pump and verify the fuel supply line for clogging. (See SUPPLY PUMP INSPECTION [SKYACTIV-D 1.5].) • Is the fuel supply line normal?	Yes	Replace the fuel injector, then go to Step 13. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Repair or replace the malfunctioning part according to the inspection results, then go to Step 13.
9	**VERIFY IF HP-EGR CONTROL VALVE MALFUNCTION IS AFFECTED** • Access the following PIDs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — HPEGR_DSD — HPEGR_ACT • Start the engine, race it at **3,000 rpm**, and idle it. • Does the monitor value of the PID item HPEGR_DSD conform to the HPEGR_ACT PID value?	Yes	Go to the next step.
		No	Inspect the HP-EGR control valve. (See HP-EGR CONTROL VALVE INSPECTION [SKYACTIV-D 1.5].) • If there is any malfunction: — Replace the HP-EGR control valve, then go to Step 13. (See HP-EGR CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • If there is no malfunction: — Go to the next step.
10	**VERIFY IF INTAKE SHUTTER VALVE MALFUNCTION IS AFFECTED** • Access the following PIDs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — ISV_POS — ISV_DSD • Start the engine, race it at **3,000 rpm**, and idle it. • Does the monitor value of the PID item ISV_POS conform to the ISV_DSD PID value?	Yes	Go to the next step.
		No	Inspect the intake shutter valve. (See INTAKE SHUTTER VALVE INSPECTION [SKYACTIV-D 1.5].) • If there is any malfunction: — Replace the intake shutter valve, then go to Step 13. (See INTAKE SHUTTER VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • If there is no malfunction: — Go to the next step.
11	**INSPECT TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY** • Inspect the turbocharger with variable turbine geometry. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to the next step.
		No	Replace the turbocharger with variable turbine geometry, then go to the next step. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

STEP	INSPECTION		ACTION
12	**INSPECT FUEL INJECTOR** • Inspect the fuel injector. (See FUEL INJECTOR INSPECTION [SKYACTIV-D 1.5].) • Are the fuel injectors normal?	Yes	Go to the next step.
		No	Replace the fuel injector in all of the cylinders, then go to the next step. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) **Note** • Even if one cylinder has a malfunction in the fuel injector, replace all of the cylinders.
13	**PERFORM FUEL INJECTOR CARBON DEPOSIT BURN-OFF PROCEDURE** **Note** • Carbon on the exhaust valve is removed by performing the fuel injector carbon deposit burn-off procedure. • Perform the "FUEL INJECTOR CARBON DEPOSIT BURN-OFF PROCEDURE". (See FUEL INJECTOR CARBON DEPOSIT BURN-OFF PROCEDURE [SKYACTIV-D 1.5].)	—	Go to the next step.
14	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Access the following PIDs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — PM_ACC_DSD — PM_GEN • Start the engine and race it at **3,000 rpm**. • Is the PM_ACC_DSD monitored value twice that of the PM_GEN monitored value or more?	Yes	Repeat the inspection from Step 2. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
15	**VERIFY AFTER REPAIR PROCEDURE** • Perform the "AFTER REPAIR PROCEDURE". (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.